

Deliverable 2 (D2)

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Project ID	
Project Title	Audient AI – An AI-Powered Conversational Intelligence Platform
Team Lead Roll Num	BITF22M002
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Capstone Project Deliverable – 2

Audient AI

“An AI-Powered Conversational Intelligence Platform”

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Table of Contents

Executive Summary	4
Design Details	4
<i>Use Case Model</i>	<i>4</i>
<i>System Use Case Diagram.....</i>	<i>10</i>
<i>Domain Model/Entity Model.....</i>	<i>11</i>
<i>ER Model (Entity-Relationship Diagram)</i>	<i>12</i>
Page Design.....	13
Case Tools.....	13

Executive Summary

Audient AI is a web-based platform that automatically captures, transcribes, translates, and summarizes professional conversations, focused on healthcare consultations. It uses AI-based speech recognition, speaker identification, translation, and smart summarization to replace manual note-taking. The system uses a React frontend, Node.js/Express backend, and PostgreSQL with AWS S3.

Four new features have been added. Offline Mode runs locally via Whisper.cpp and Ollama with no internet required. Vocal Prompts enable hands-free control through spoken commands like "Start Recording." English Scripting standardizes all transcription output to English regardless of source language. AI Field Reminders alert users when critical fields like diagnosis or medication are left incomplete before finalizing. The work plan covers analysis, design, prototype, evaluation, and deployment.

Design Details

This section presents the comprehensive system design for Audient AI using industry-standard Unified Modeling Language (UML) diagrams and data models. The design captures system behavior, structure, and data organization at multiple levels of abstraction.

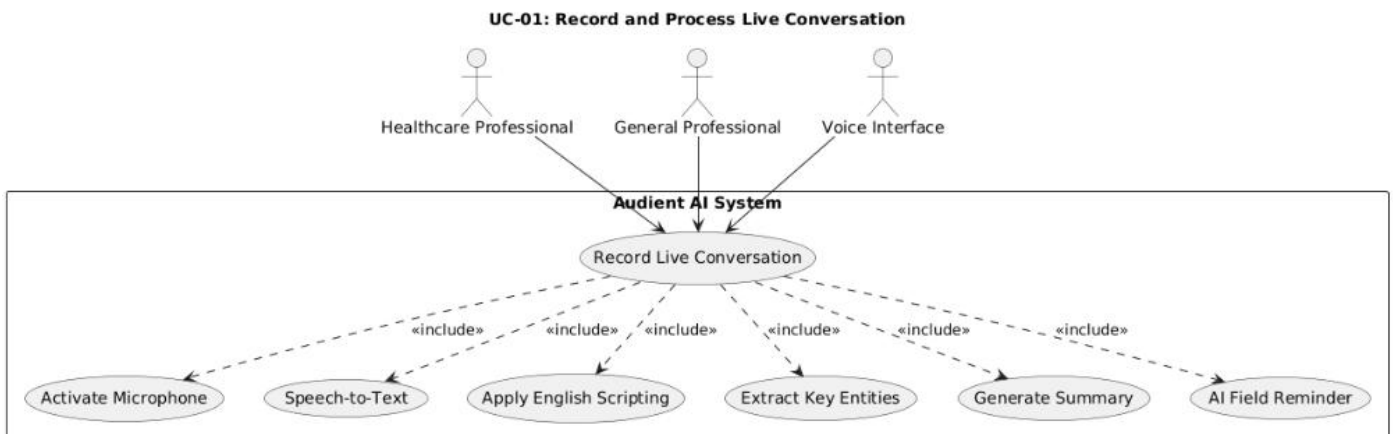
Use Case Model

Actors:

Actor	Description
Healthcare Professional	Primary user who records patient consultations and accesses medical documentation summaries
General Professional User	Business professionals, legal practitioners who record meetings and discussions
System Administrator	Manages user accounts, system configuration, and monitors platform performance
External AI Services	Third-party APIs (OpenAI Whisper, GPT, Google Translate) that process audio and text
Offline LLM Engine	Locally deployed models (Whisper.cpp, Ollama) that handle speech recognition and summarization without internet connectivity in offline mode
Voice Interface	Vocal prompt subsystem that listens for trigger commands (e.g., 'Start Recording', 'Stop Recording') and translates them into system actions

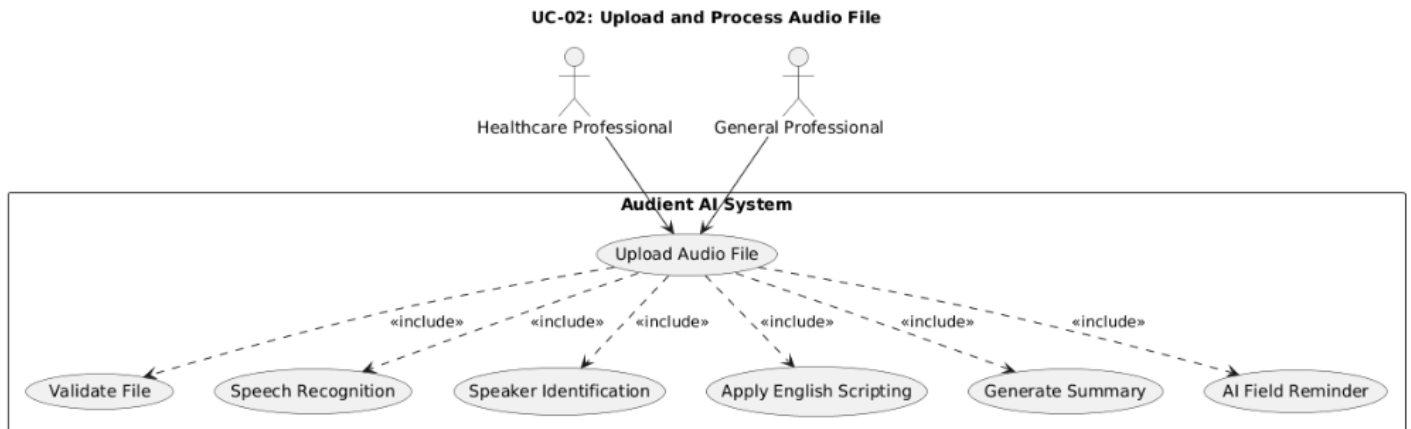
Main Use Cases:

Use Case 1: Record and Process Live Conversation



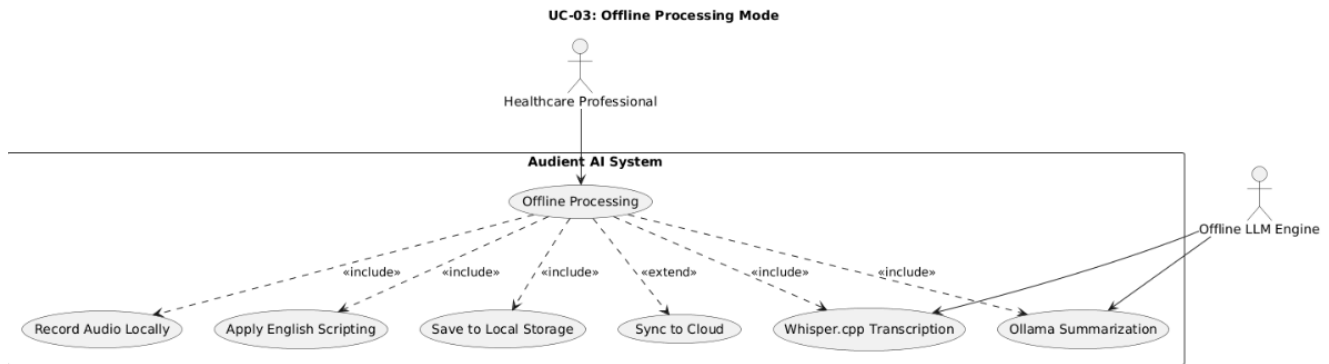
Field	Detail
ID	UC-01
Actors	Healthcare Professional / General Professional / Voice Interface
Preconditions	User authenticated and logged in
Main Flow	1. User starts recording via button OR vocal command "Start Recording" 2. System activates mic; Voice Interface confirms audibly 3. User conducts conversation 4. User stops via button OR vocal command "Stop Recording" 5. Audio uploaded to AWS S3 (online) OR stored locally (offline) 6. Speech-to-Text via OpenAI Whisper (online) OR Whisper.cpp (offline) 7. System receives transcript with speaker labels 8. English Scripting normalizes all output to English 9. System extracts entities (names, dates, diagnoses, medications) 10. AI generates intelligent summary 11. AI Field Reminder checks all required/optional fields 12. Final summary displayed to user
Postconditions	Summary stored in dashboard; all fields validated

Use Case 2: Upload and Process Audio File:



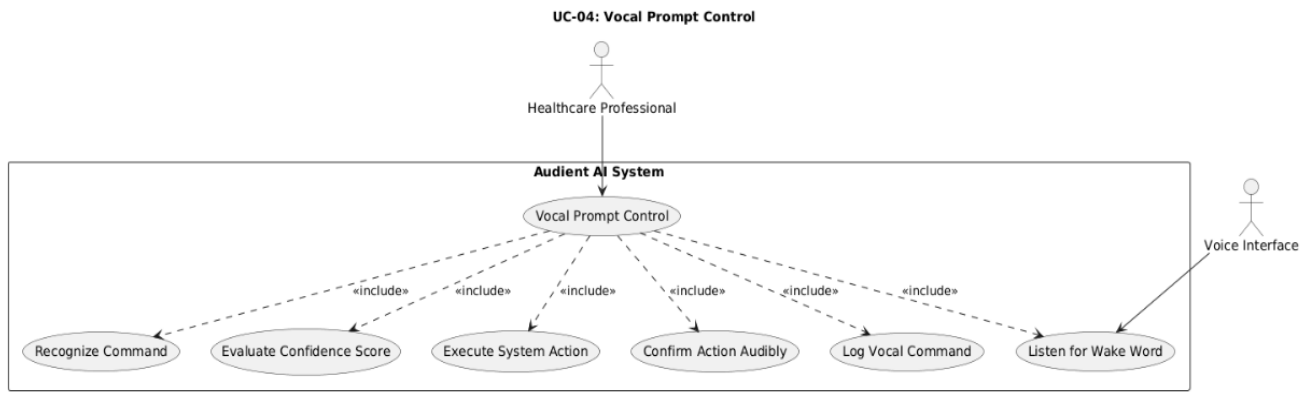
Field	Detail
ID	UC-02
Actors	Healthcare Professional / General Professional
Preconditions	User has audio file in supported format (MP3, WAV, M4A)
Main Flow	1. User clicks "Upload Audio" 2. Selects file from device 3. System validates format and size 4. File uploaded to AWS S3 5. Speech recognition processes audio 6. System performs speaker identification 7. English Scripting applied to transcript 8. Multilingual content translated 9. Key entities extracted 10. Summary generated 11. AI Field Reminder checks incomplete fields 12. Summary displayed to user
Postconditions	Audio file and summary stored in user account

Use Case 3 Offline Processing Mode



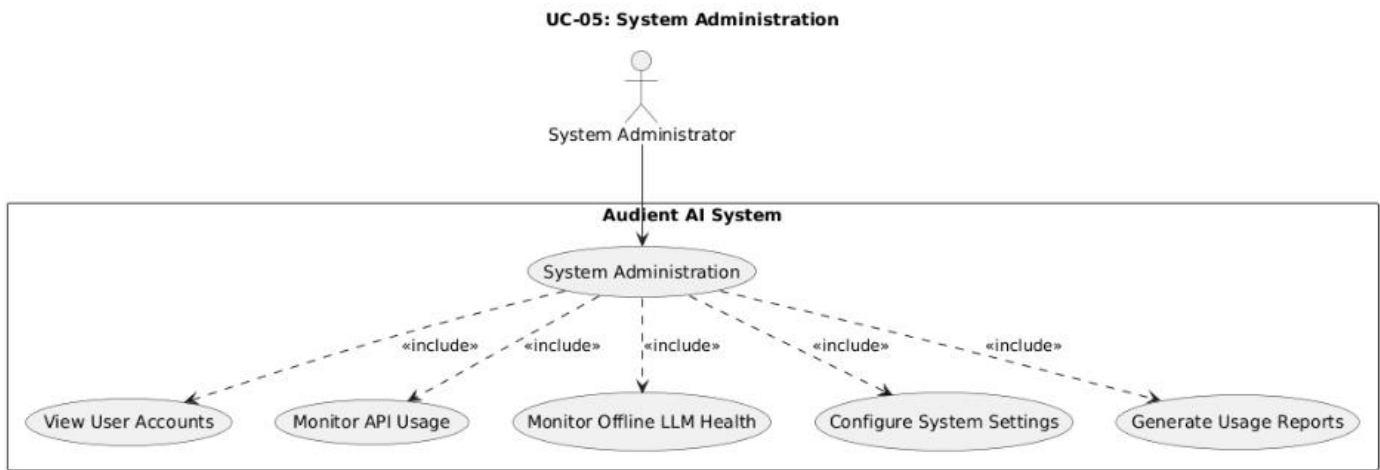
Field	Detail
ID	UC-03
Actors	Healthcare Professional / Offline LLM Engine
Preconditions	Internet unavailable OR user selected Offline Mode in settings
Main Flow	1. System detects offline state OR user manually selects Offline Mode 2. Recording begins using local microphone 3. Audio processed by Whisper.cpp for speech-to-text 4. Transcript passed to Ollama (Mistral/LLaMA 3) for summarization 5. English Scripting and entity extraction run locally 6. Summary saved to local storage with sync_status = pending 7. When internet restored, system auto-syncs session to cloud 8. Sync status updated to synced
Postconditions	Session stored locally and synced to cloud when connectivity restored

Use Case 4: Vocal Prompt Control



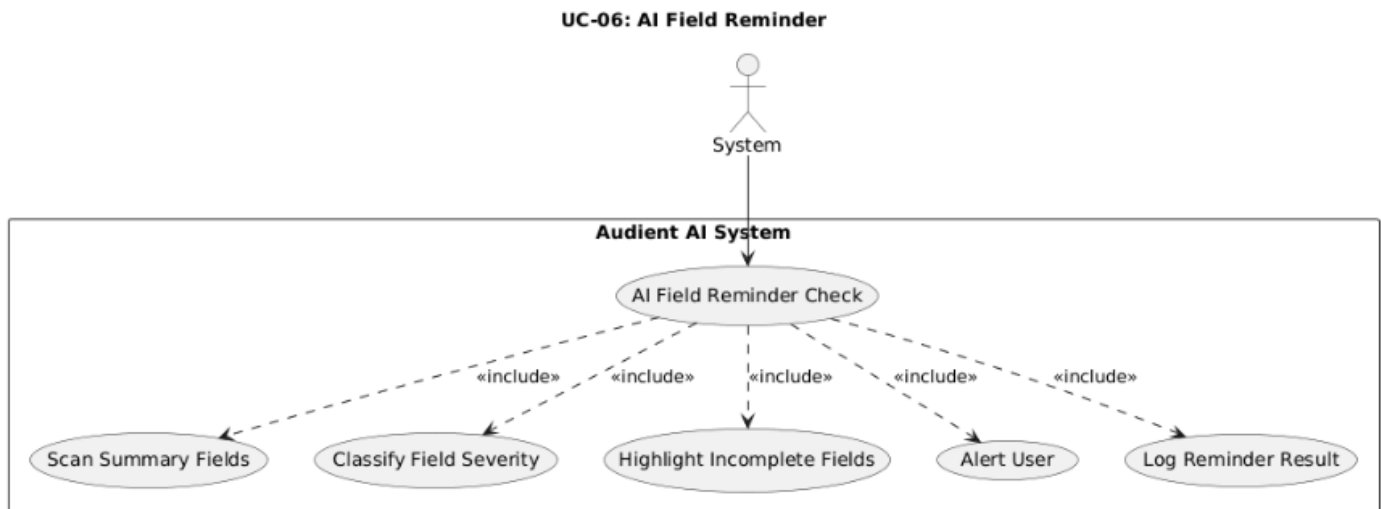
Field	Detail
ID	UC-04
Actors	Healthcare Professional / Voice Interface
Preconditions	Vocal prompts enabled in settings; microphone active
Main Flow	1. System continuously listens for configured wake word 2. User speaks command ("Start Recording", "Stop Recording", "Generate Summary") 3. Voice Interface processes audio and identifies command 4. Confidence score evaluated against threshold 5. Matched command mapped to system action 6. System executes action immediately 7. Voice Interface confirms action audibly to user 8. Command logged in VOCAL_COMMANDS table with timestamp
Postconditions	System action executed; vocal command recorded in database

Use Case 5: System Administration:



Field	Detail
ID	UC-05
Actors	System Administrator
Preconditions	Administrator logged in with admin privileges
Main Flow	1. Admin accesses admin dashboard 2. System displays platform-wide statistics 3. Admin views and manages all user accounts 4. Admin monitors API usage and associated costs 5. Admin monitors offline LLM model health and performance 6. Admin configures system-wide settings 7. Admin handles user support requests 8. Admin generates and exports usage reports
Postconditions	System configuration updated or reports generated

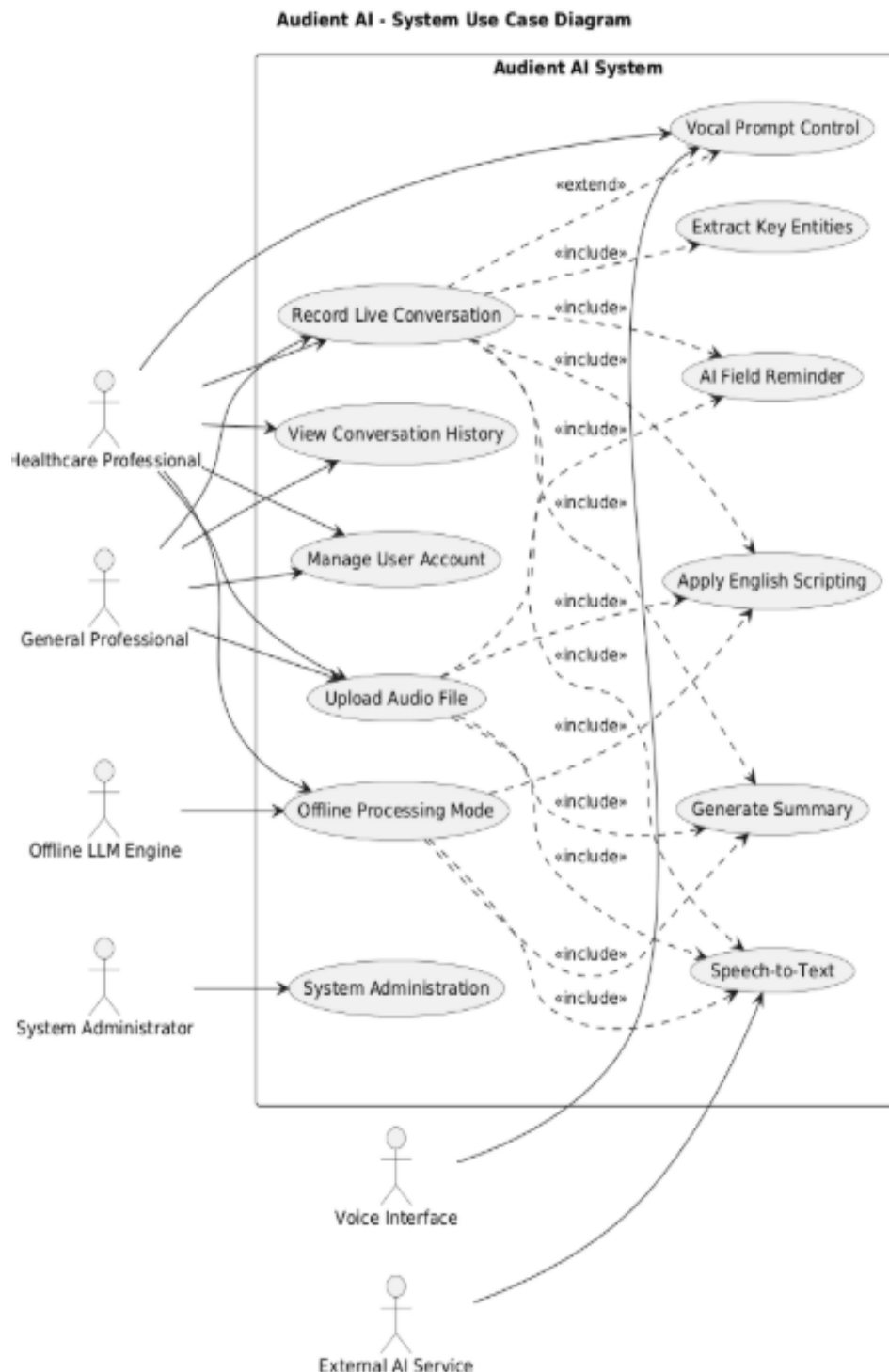
UC-06: AI Field Reminder



Field	Detail
ID	UC-06
Actors	System (auto-triggered after summary generation)
Preconditions	Summary has been generated; session not yet finalized
Main Flow	1. System automatically triggers Field Reminder after summary generation 2. AI scans all summary form fields 3. Each field classified as critical, important, or optional 4. Missing or incomplete fields identified 5. System highlights fields with color-coded severity (red/yellow/grey) 6. User alerted with inline notification 7. User fills missing fields OR confirms override 8. Result logged in FIELD_REMINDERS table with resolution status
Postconditions	All field statuses recorded; session finalized only after user confirmation

System Use Case Diagram:

The system use case diagram has been updated to reflect the addition of four new features. The End User actor now connects to UC-07 (Vocal Prompt Control) in addition to all prior use cases. The Offline LLM Engine actor connects to UC-06 (Offline Processing Mode) and to the Process Audio pipeline. The Voice Interface actor connects to UC-07. UC-08 (AI Field Reminder) is invoked automatically by the system after any audio processing use case is completed. English Scripting is represented as an include relationship within the Process Audio flow, alongside Translation, Speech-to-Text, Speaker Identification, and Entity Extraction.

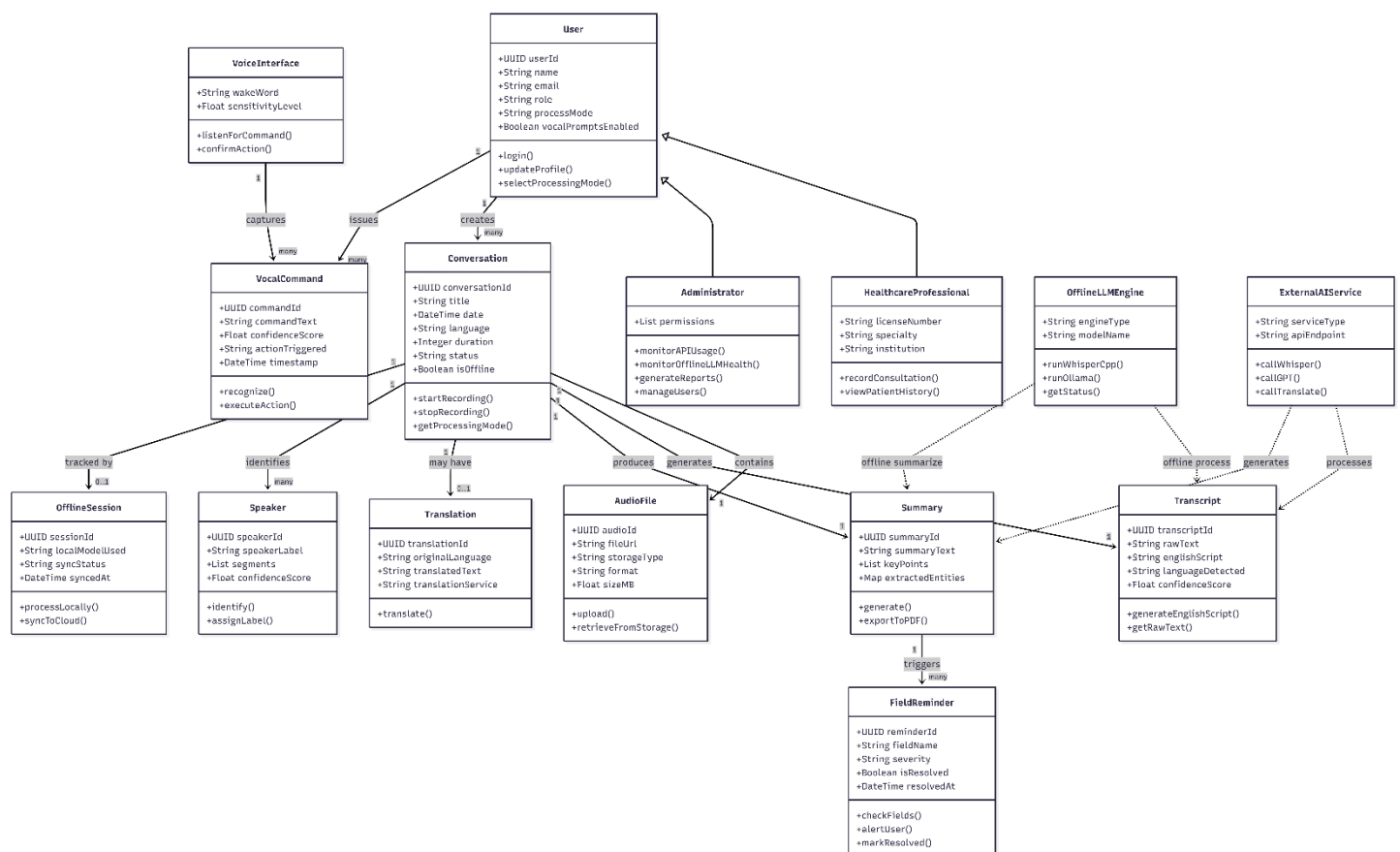


Domain Model/Entity Model:

The domain model represents the core business entities of the Audient AI system, identified using the noun-hunting technique from project requirements.

- The **User entity** serves as the base class containing credentials and profile information, following the Information Expert GRASP pattern. It specializes into HealthcareProfessional (with medical licensing attributes) and Administrator (with system management permissions) through inheritance relationships. A new processMode attribute (online/offline) is added to User to store the preferred processing mode.
- The **Conversation entity** acts as the central aggregate root representing a recording session. Following the Creator pattern, each user can create multiple conversations. Every conversation maintains one-to-one relationships with AudioFile (storing recording metadata and AWS S3 or local path references), Transcript (containing speech-to-text output), and Summary (holding AI-generated summaries with extracted entities in JSON format). A new englishScript attribute on Transcript records the English-normalized version of the transcription.
- The **Translation entity** provides optional multilingual support, storing converted text when needed. The Speaker entity enables speaker identification by maintaining labels and timestamp segments for multiple participants per conversation.

Three new entities are introduced. The VocalCommand entity records each voice command issued by a user, including the raw audio snippet, recognized text, confidence score, and the action triggered. The OfflineSession entity tracks conversations processed in offline mode, storing the local model used, sync status, and sync timestamp for later cloud reconciliation. The FieldReminder entity stores the result of each AI reminder check, recording which fields were flagged, their severity (critical / important / optional), whether they were resolved, and the resolution timestamp.



ER Model (Entity-Relationship Diagram)

The Entity-Relationship diagram shows the normalized database structure using PostgreSQL for relational data and AWS S3 (or local filesystem in offline mode) for audio storage.

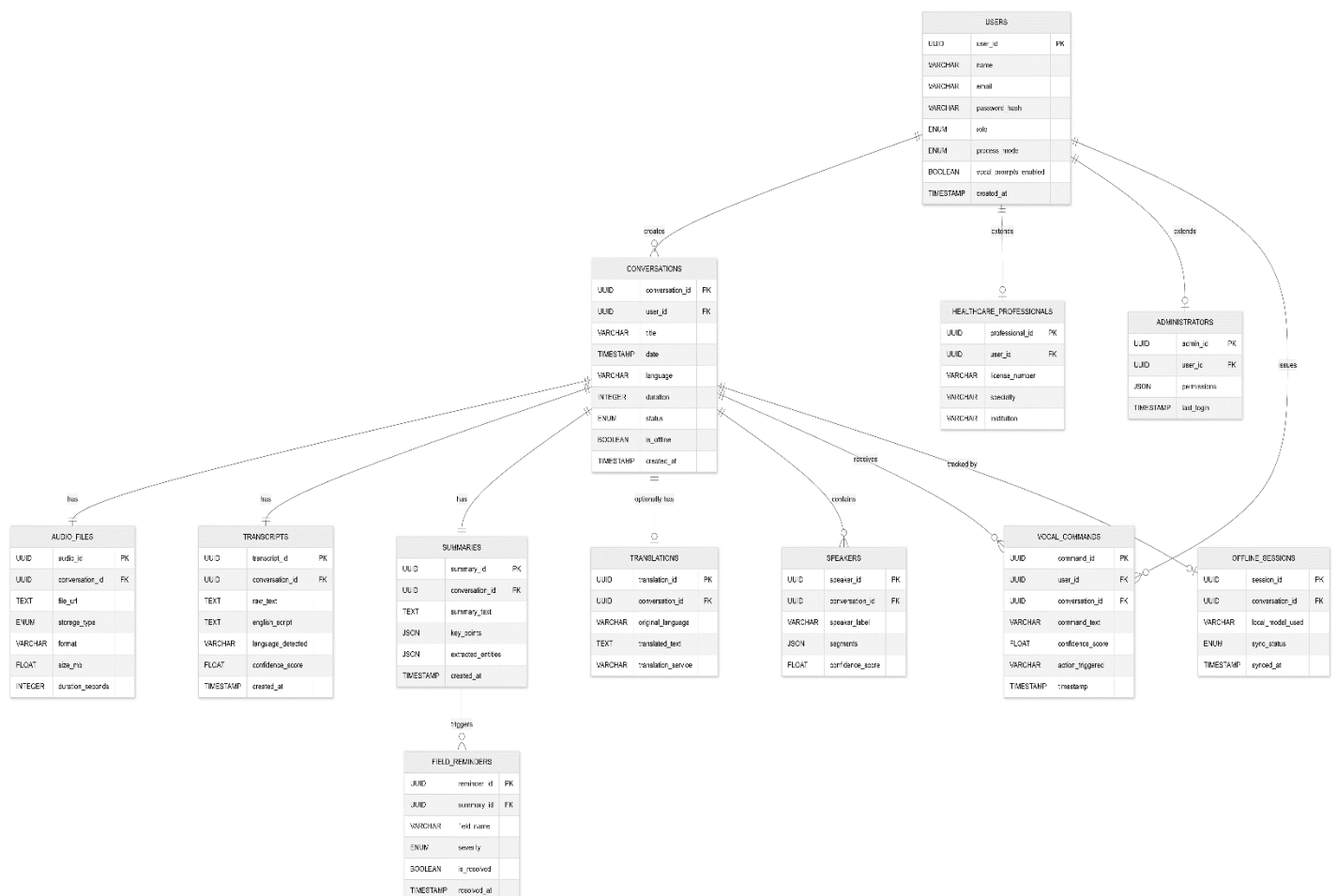
User Management: The USERS table stores common user information, including a new process_mode column (ENUM: 'online', 'offline') and a vocal_prompts_enabled boolean flag. HEALTHCARE_PROFESSIONALS and ADMINISTRATORS tables extend USERS through foreign key relationships, implementing inheritance at the database level.

Conversation Management: The CONVERSATIONS table is the central entity with a one-to-many relationship to USERS. It stores metadata including date, language, processing status, duration, and a new is_offline boolean to record whether the session was processed in offline mode.

Data Processing: Each conversation has mandatory one-to-one relationships with AUDIO_FILES (AWS S3 URLs or local filesystem paths and metadata), TRANSCRIPTS (speech-to-text output, including a new english_script TEXT column for the English-normalized transcript), and SUMMARIES (AI summaries with key points and extracted entities stored as JSON). These relationships use unique foreign key constraints.

Additional Features: TRANSLATIONS table provides optional one-to-one support for multilingual conversations. SPEAKERS table maintains one-to-many relationships for speaker identification with labels, segments, and confidence scores.

New Tables: VOCAL_COMMANDS table stores each recognized vocal command with columns for user_id, conversation_id, command_text, confidence_score, action_triggered, and timestamp. OFFLINE_SESSIONS table tracks conversations processed offline with columns for conversation_id, local_model_used, sync_status (ENUM: 'pending', 'synced', 'failed'), and synced_at timestamp. FIELD_REMINDERS table stores AI reminder check results with columns for summary_id, field_name, severity (ENUM: 'critical', 'important', 'optional'), is_resolved boolean, and resolved_at timestamp.



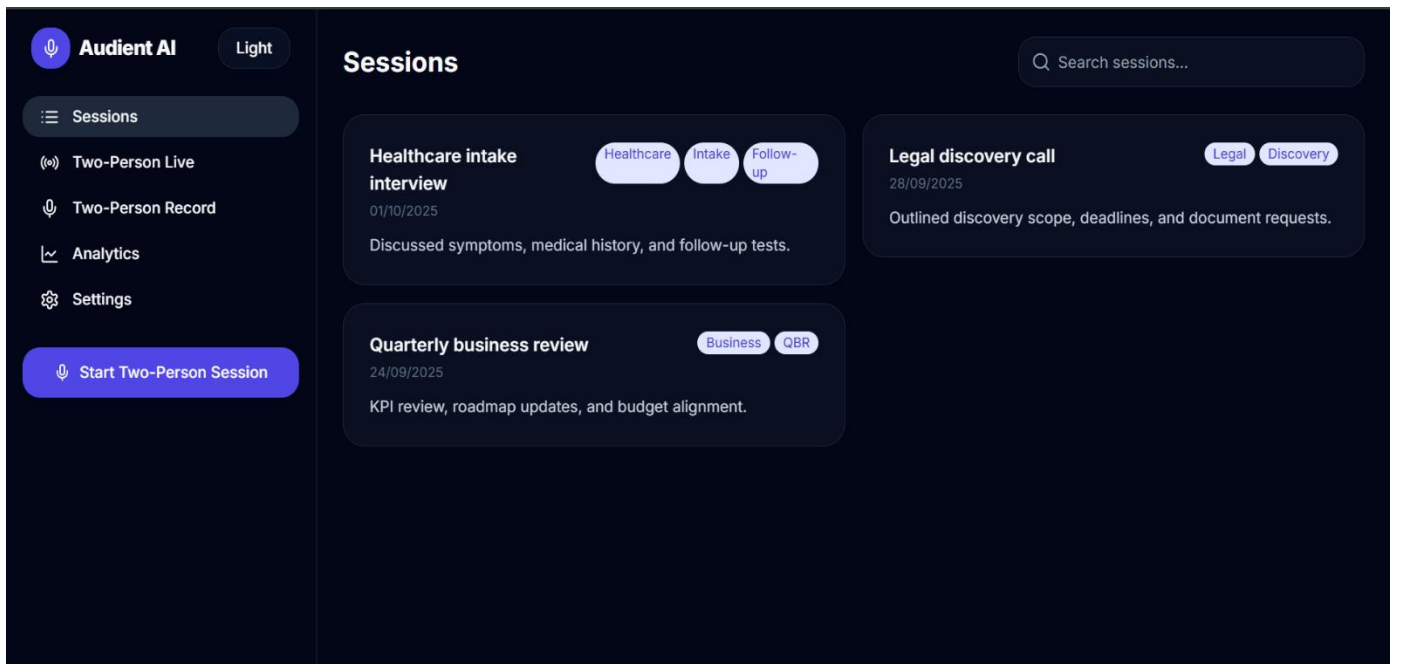
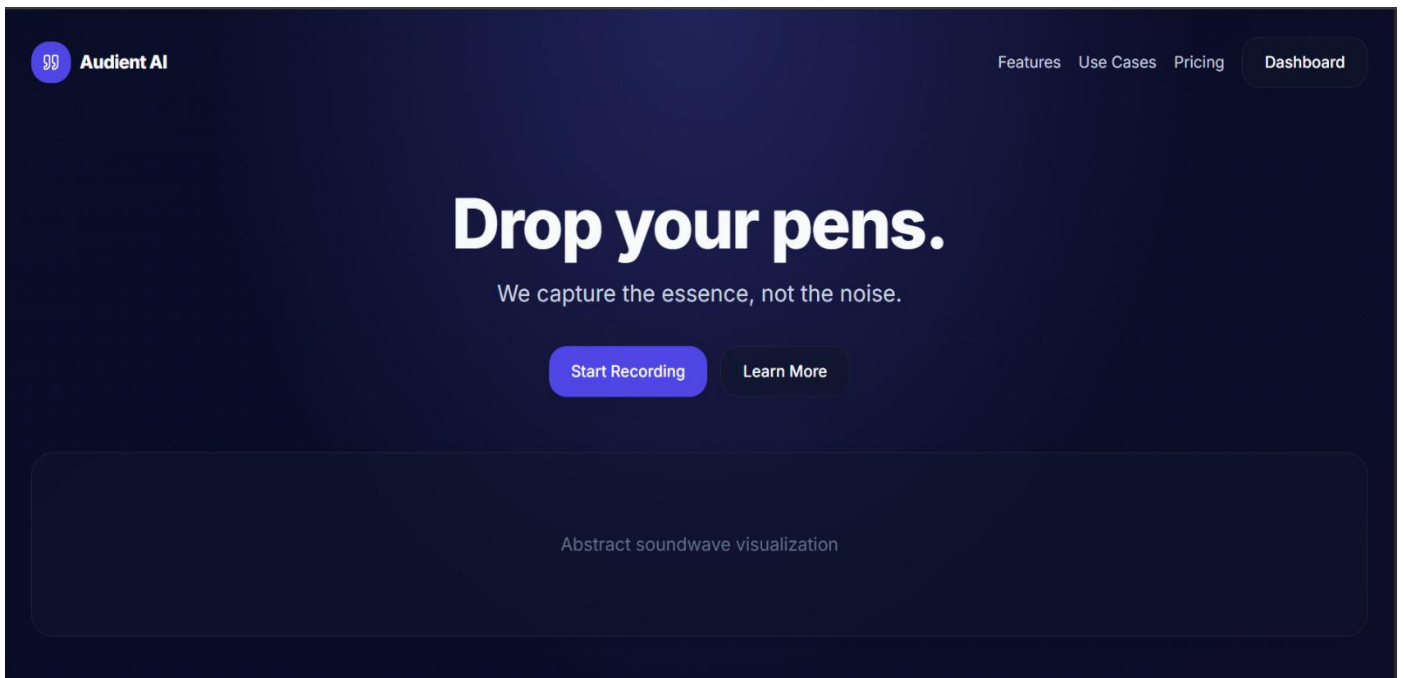
Case Tools

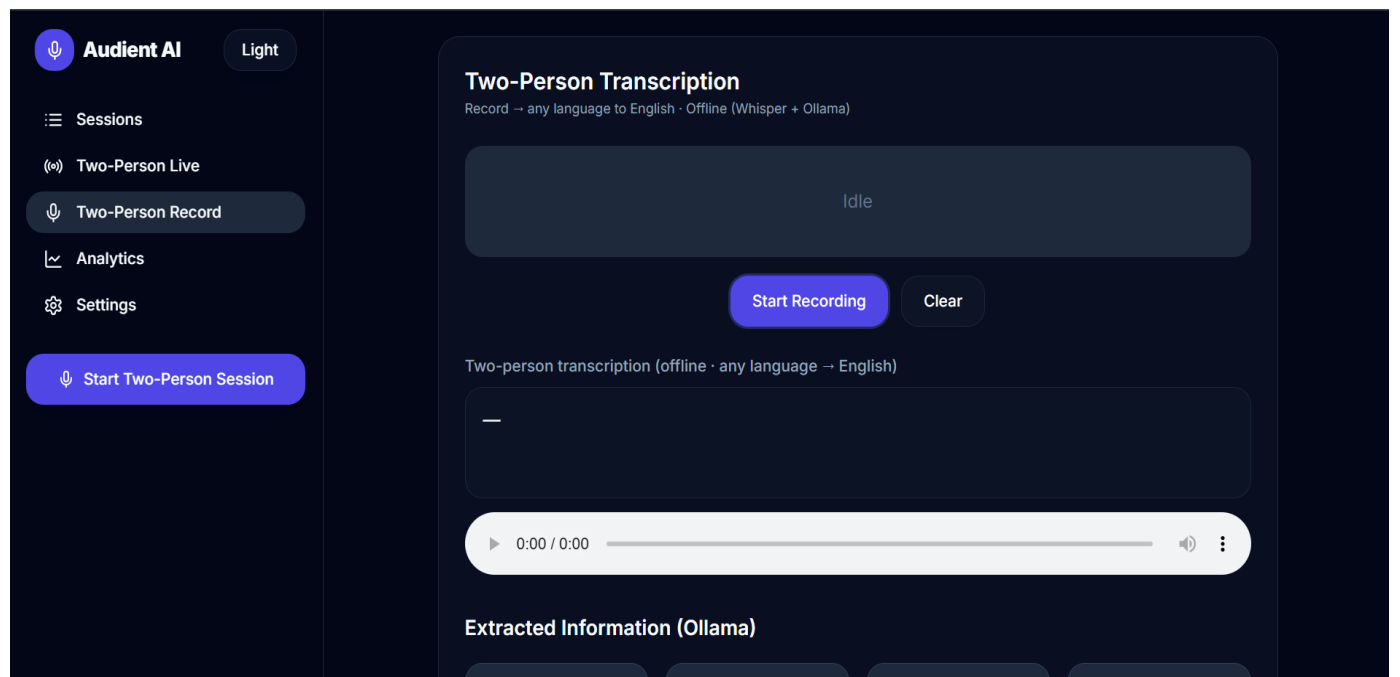
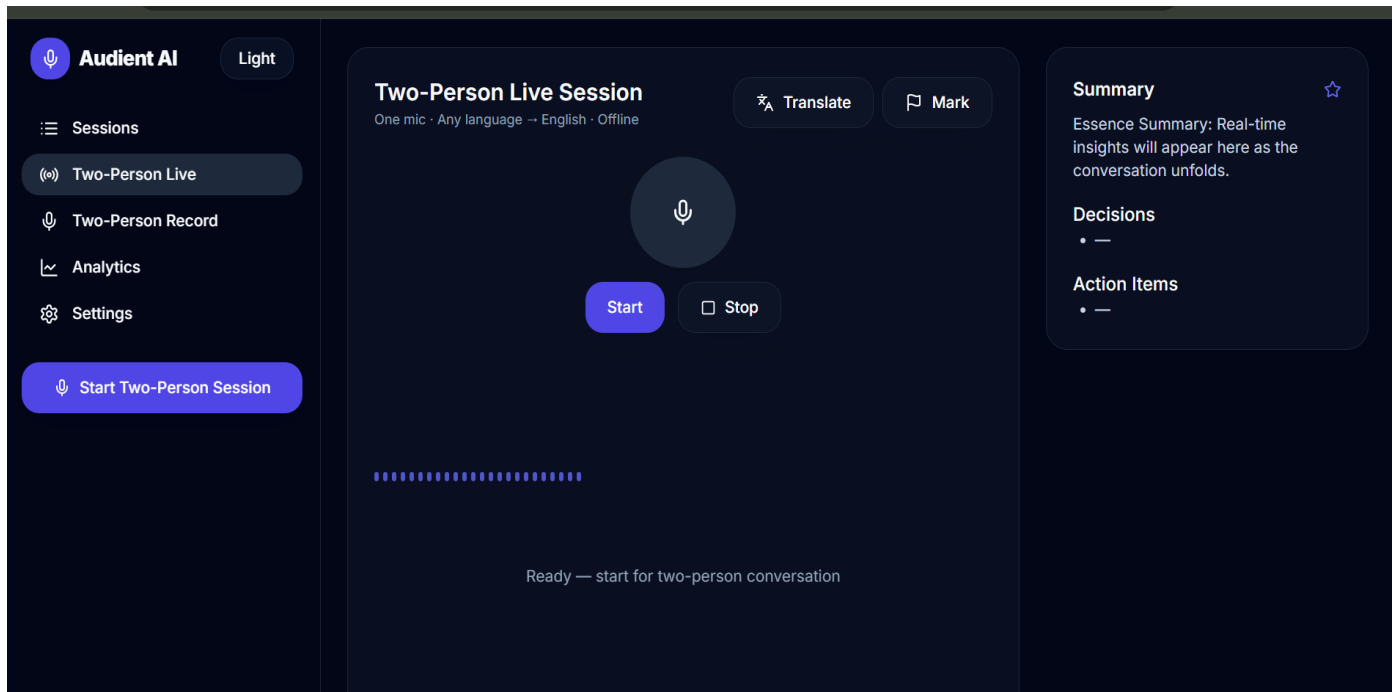
Category	Tool	Reason
UML Modeling	Draw.io / PlantUML	Free, UML support, easy integration
UI/UX Mockups	Figma	Fast wireframing and collaboration
Project Management	Trello, Jira	Simple task tracking and milestones
Documentation	MS Word	Standard academic documentation
Repository	GitHub	Version control and collaboration
API Testing	Postman	Backend API testing
Offline STT	Whisper.cpp	Local on-device speech-to-text for offline mode; no internet required
Offline LLM	Ollama (Mistral / LLaMA 3)	Local LLM serving for offline summarization and entity extraction
Voice Processing	Web Speech API / Vosk	Browser-native or offline library for vocal prompt command recognition

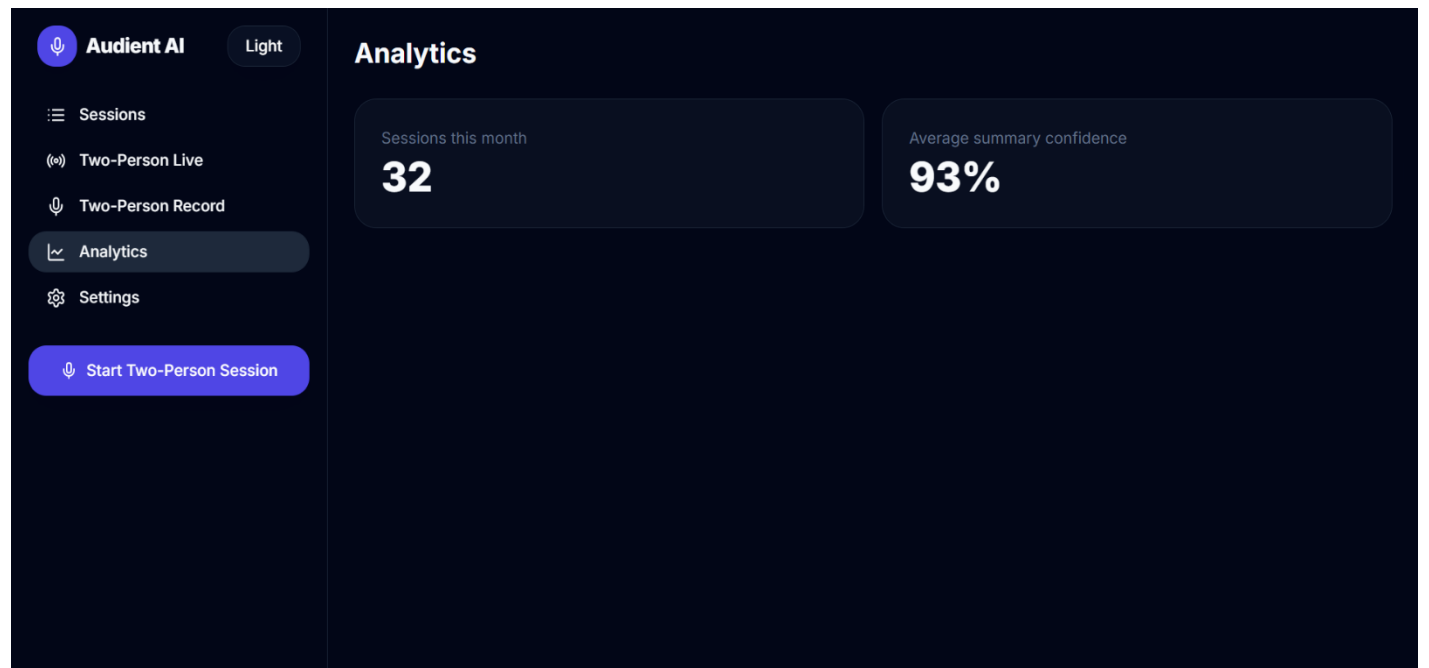
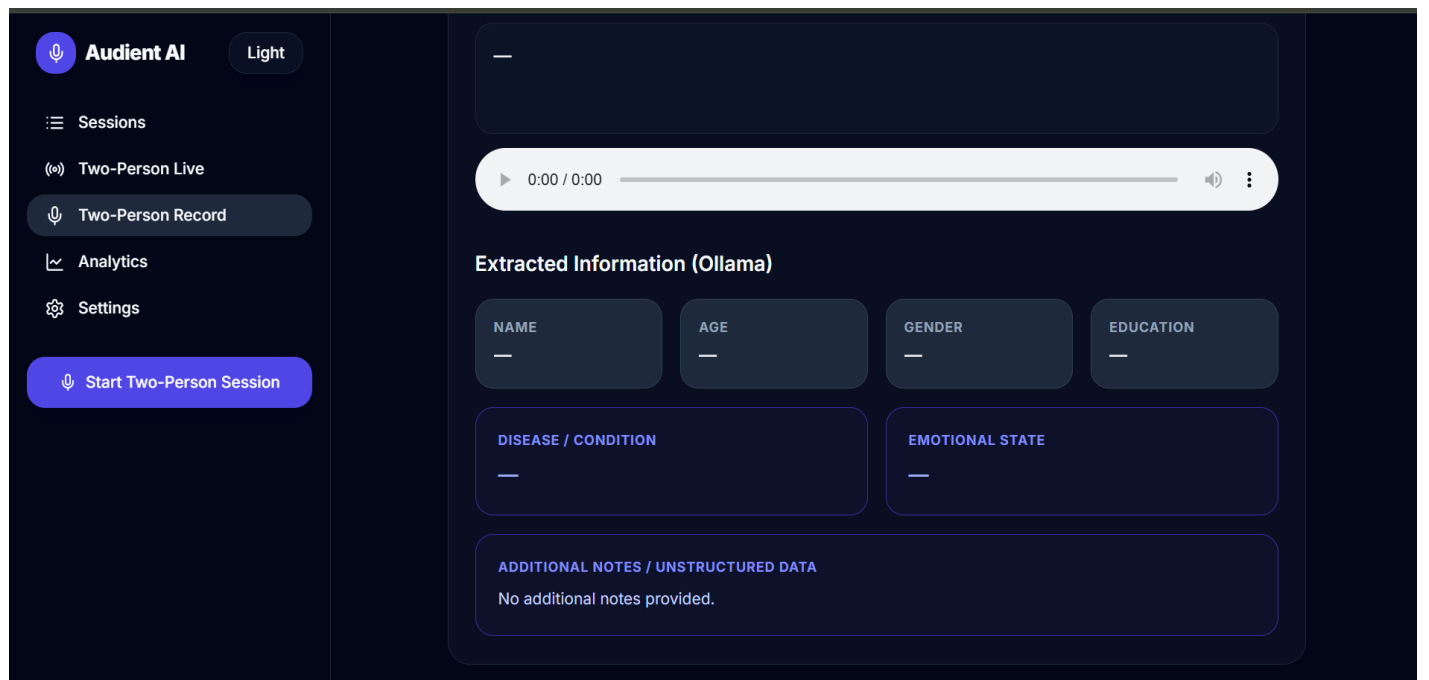
Appendix – CASE Tool Justification

UML CASE tools such as Draw.io and PlantUML are selected due to free availability and compatibility with development environments. Trello is selected for lightweight project tracking without heavy framework overhead. MS Word is used for formal documentation. GitHub is used for repository and version control with strong community and integration support. These tools satisfy low-cost, integration-friendly, and learning-support requirements.

Page Design







Bright Mode:

Two-Person Live Session

One mic · Any language → English · Offline

 Translate

 Mark



Start

☐ Stop

.....

Ready — start for two-person conversation

Summary

Essence Summary: Real-time insights will appear here as the conversation unfolds.

Decisions

• —

Action Items

• —

